Chapter 5, Problem 52P

Problem

Design an op amp circuit such that

vo = 4v1 + 6v2 − 3v3 − 5v4

Let all the resistors be in the range of 20 to 200 kΩ.

Step-by-step solution

Step 1 of 4

The output voltage of an op amp circuit is

$v_o = 4v_1 + 6v_2 - 3v_3 - 5v_4$

Let the range of all the resistors $= 20\text{ to }200\,\text{k}\Omega$

We can achieve this output with a summing amplifier with two inverters for inverting v_1 and v_2.

Draw the circuit diagram as shown in Figure 1.

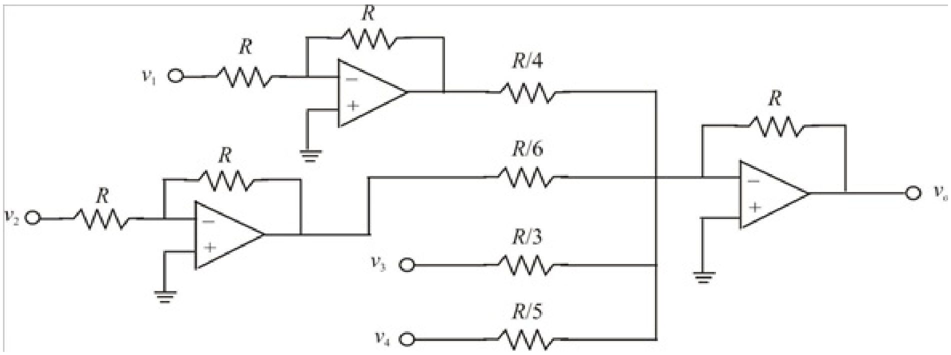

Figure 1

Step 2 of 4

Let the resistance $R = 36\,\text{k}\Omega$

Then the desired circuit is shown in Figure 2.

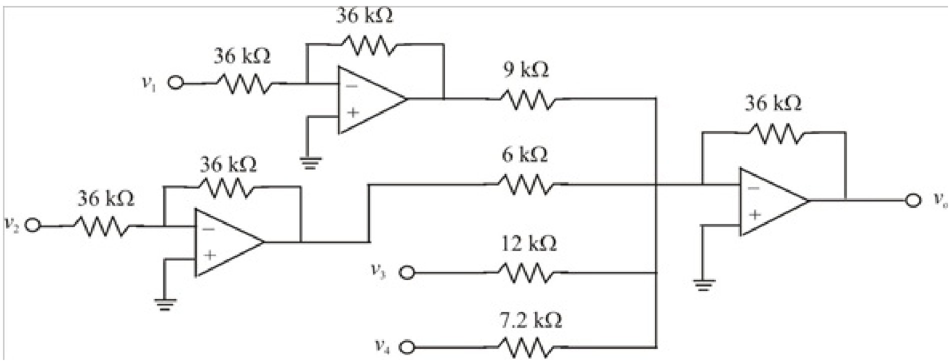

Figure 2

Step 3 of 4

The output voltage of inverting amplifier is given by:

$$v_o = -\frac{R_f}{R_1} v_i$$

The output voltage v_a of inverter is calculated as:

$$\begin{aligned} v_a &= -\frac{R_f}{R_1} v_1 \\ &= -\frac{36}{36}(v_1) \\ &= -v_1 \end{aligned}$$

The output voltage v_b of inverter is calculated as:

$$\begin{aligned} v_b &= -\frac{R_f}{R_1} v_2 \\ &= -\frac{36}{36}(v_2) \\ &= -v_2 \end{aligned}$$

Step 4 of 4

The output voltage of a four-input summer is given by:

$$v_o = -\left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3 + \frac{R_f}{R_4} v_4\right)$$

From the Figure 2, the output voltage of a four-input summer is calculated as:

$$\begin{aligned} v_o &= -\left(\frac{R}{R/4} v_a + \frac{R}{R/6} v_b + \frac{R}{R/3} v_3 + \frac{R}{R/5} v_4\right) \\ &= -\left(4v_a + 6v_b + 3v_3 + 5v_4\right) \end{aligned}$$

Therefore $v_o = -\left(4v_a + 6v_b + 3v_3 + 5v_4\right)$ …… (1)

But,

$v_a = -v_1$

and $v_b = -v_2$

Substituting these values in equation (1), we get

$$\begin{aligned} v_o &= -\left(4v_a + 6v_b + 3v_3 + 5v_4\right) \\ &= -\left(-4v_1 - 6v_2 + 3v_3 + 5v_4\right) \\ &= 4v_1 + 6v_2 - 3v_3 - 5v_4 \end{aligned}$$

Therefore the circuit in Figure 2 achieves the output voltage is $\boxed{v_o = 4v_1 + 6v_2 - 3v_3 - 5v_4}$.